



Machine learning driven metaheuristics to optimize the design-operational parameters for enhanced pool boiling CHF subject to a range of surface-liquid combinations

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ABSTRACT

Critical heat flux (CHF) is crucial for heat-flux controlled boiling applications. Designing heat sinks with larger CHF limits is essential for two-phase thermal systems. However, identifying optimal operational and heater surface morphological parameters remains challenging. Most empirical correlations focus on CHF prediction rather than inverse design, showing large errors and limited applicability. We previously developed a deep neural network (DNN) for CHF using a database of 13,000 data points from 42 pool boiling studies, incorporating 29 working fluids across various surface textures and operating conditions. The DNN model achieved a R^2 score of 0.97. The present study addresses the inverse design problem: identifying optimal design and operational parameters for targeted CHF values. An optimized Genetic Algorithm (GA) model is proposed to predict these parameters. The analysis includes HFE-7000, Water, HFE-7200, and FC-72, evaluated under typical CHF benchmarks for substrates with thermal conductivities from 205 to 385 W/m.K. For HFE-7000, at CHF of 120 kW/m² and 252 kW/m² on a substrate with 385 W/m.K conductivity, inclination angle values decrease as CHF increases, due to vapor removal difficulties. For HFE-7200 at CHF of 41.11 kW/m² and 93.8 kW/m² on a 205 W/m.K substrate, parameter values increase with CHF, except temperature and surface roughness. These models serve as tools for predicting CHF and identifying optimal parameters for heating surfaces to meet desired CHF targets.

1. Introduction

Over the past four decades, several technologies have achieved surprising improvements in heat dissipation rates, while the preference for smaller and lighter system architectures has also increased [1–3]. This enormous heat cannot be tackled by single phase systems and thus two phase systems are the prime candidates for such applications [2,4,5]. Among those, pool boiling is not only the most widely used and developed two-phase cooling method in industry, but it is also the simplest and most affordable [6–10]. Examples of the aforementioned are cooling of electronics components, power applications, and superconductor

coils, where pool boiling makes use of the coolant's sensible and latent heat to accommodate large-scale heat removal while maintaining system temperatures conveniently below ranges governed by both substance and system reliability constraints [11,12]. Heat transfer coefficient ($q/\Delta T$) and CHF are the key design and performance parameters of the pool boiling heat transfer [13,14]. For heat flux controlled cooling applications, surpassing CHF results in a quick and erratic switch from highly efficient nucleate boiling to extremely heat transfer inefficient film boiling. Moreover, there is a sudden rise in surface temperature that can cause physical harm, melting, or burnout of the system [15].

Abbreviations: ANN, Artificial Neural Network; CHF, Critical Heat Flux; DNN, Deep Neural Network; GA, Genetic Algorithm; HTC, Heat Transfer Coefficient; MAE, Mean Absolute Error; ML, Machine Learning; PINN, Physics-Informed Neural Network; R^2 , Coefficient of Determination (R-squared); XAI, Explainable Artificial Intelligence.

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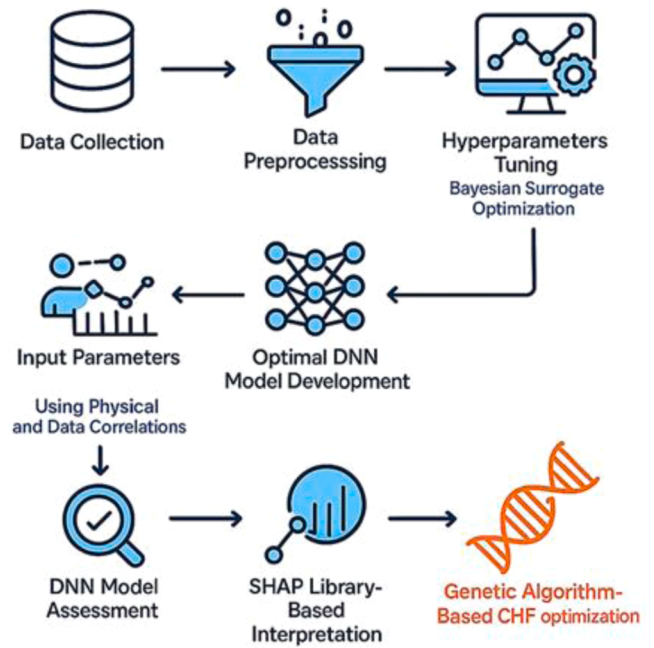
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Table 1

Collected pool boiling data on roughened surfaces reported in literature including working fluid.

Previous Work	Working Fluid
Kim et al. [46]	FC-72
Al Masri et al. [47]	Acetone
Tran et al. [48]	HFE-7200
McHale and Garimella [49]	FC-77
Kim et al. [50]	Water
Bombardieri and Manfretti [51]	Liquid Nitrogen
Bon et al. [52]	Pentane and Butane
Alvarino et al. [53]	HFE-7100
Ahmad et al. [54]	R-123
Fan et al. [55]	Water
Arenales et al. [56]	Water
Suszko and El-Genk [57]	PF-5060
Thiagarajan et al. [58]	HFE-7100
Kim et al. [59]	Water
El-Genk and Pourghasemi [60]	HFE-7000
McHale and Garimella [61]	FC-72
McHale and Garimella [62]	FC-72
Kruse et al. [63]	DI Water
Ribatski and Jabardo [64]	R-11, R-123, R-12, R-134a, and R-22
Benjamin and Balakrishnan [65]	Distilled Water, n-hexane, Acetone and CCl ₄
Bock et al. [66]	R-245fa
Bock et al. [67]	R-134a
Bode [68]	Water
Eid et al. [69]	R-141b
Ferjančić and Golobič [70]	FC-72 and Water
Goel et al. [71]	Demineralized Water
Jabardo et al. [72]	R-134 and R-123
Kang [73]	Water
Kim et al. [74]	Water
Kim et al. [75]	DI water
Kiyomura et al. [76]	FC-72, FC-87, DI Water and n-Pentane
Luke [77]	Propane
Luke [78]	Propane
Mpholo et al. [79]	Water
Parker and El-Genk [80]	FC-72
Pioro et al. [81]	Freon-113, Water, R-11
Qi et al. [82]	Water
Sarker et al. [83]	DI Water
Suszko and El-Genk [84]	PF-5060
Swain et al. [85]	Isopropyl Alcohol, Acetone and Ethyl Alcohol
Walunj and Sathyabhama [86]	Water
Zhang et al. [87]	Water

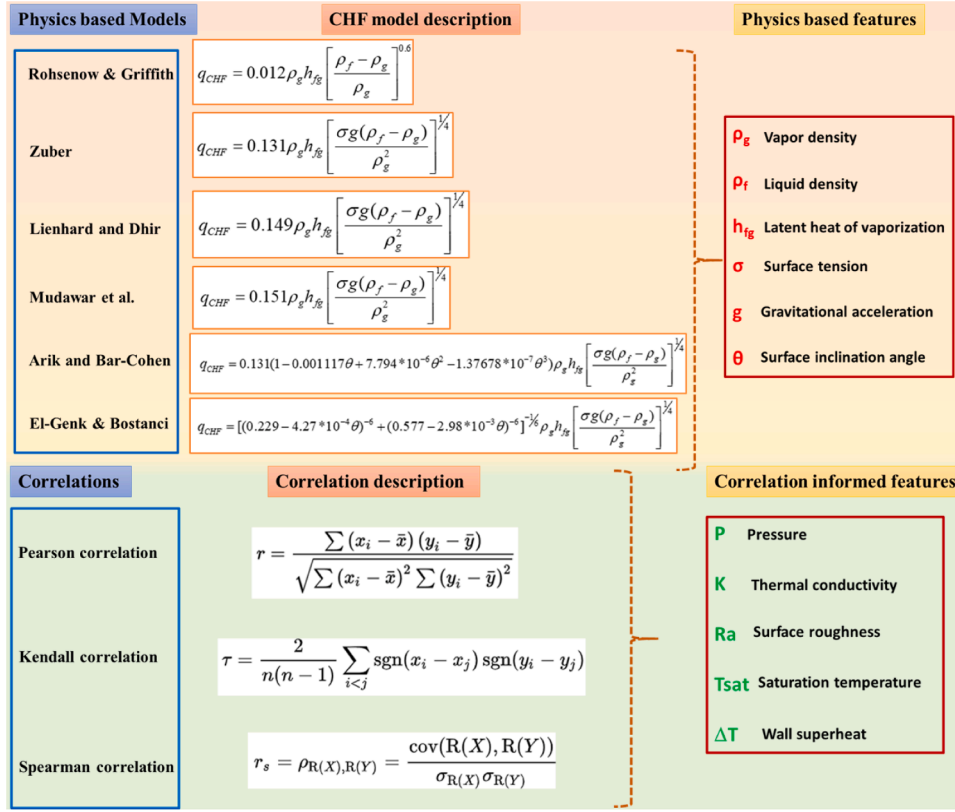
Therefore, it is considered as a bottleneck for thermal systems and in the past few decades, various efforts have been made to delay or extend the CHF limit by varying combination of surface, working fluid, and operating conditions. For instance, Kim and Kim [16] investigated enhancement in CHF for the nanoparticle (NiCr) coated surface by using three nanofluids (TiO₂, Al₂O₃, SiO₂). They found an almost 3-fold increase in CHF by adding 0.1 % concentration of nanofluid SiO₂ as compared to pure water. Similarly, Chang and You [17] examined the pool boiling of FC-72 on diamond-particle-coated microporous/porous surfaces. They noted a significant enhancement in CHF as compared to the bare surface. Hwang and Kaviany [18] investigated the pool boiling of n-pentane on thin, uniform coating of particle diameter ranging between 40 and 80 μm with the coating thickness of 3 to 5 particle layers on the copper surface. They noted around 1.8 times enhancement in CHF as compared to bare surface. They suggested that thin, porous coating was able to decrease and increase the critical Rayleigh-Taylor and vapor channel area, respectively. Byon et al. [19] experimentally investigated the pool boiling of FC-72 on sintered coated copper surfaces having particle diameter and coating thickness between 45 and 100 μm and 0.7 to 3 mm, respectively. They found that CHF for bi-porous coatings was 72 % higher than that of nanoporous coatings. Based on experimental data they developed a correlation to predict CHF as a function of particle to cluster size ratio for the biporous coated surfaces. Chu et al. [20] investigated the pool boiling of water on the microstructure rough surfaces and obtained a CHF of 208 W/cm²(about 160 % increase as

**Fig. 1.** An overview of the methodological steps for CHF optimization.

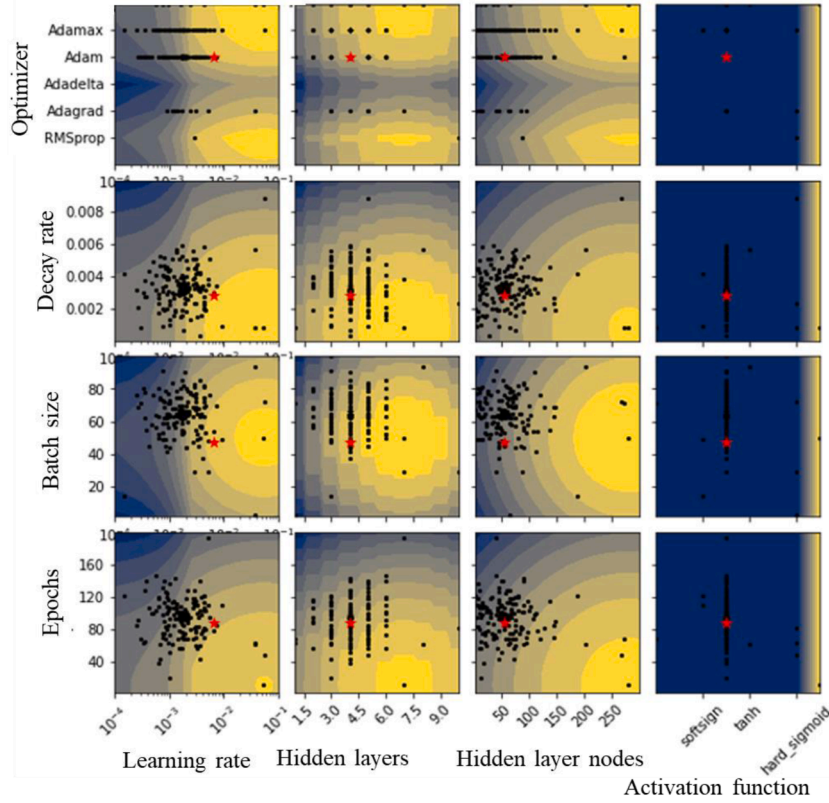
compared to reference untreated surface) from the surface having maximum roughness(ratio of true area in contact with liquid to the projected area) of around 6. They noted a clear trend of increase in CHF with the increase in roughness which they attributed to the increase in capillary force due to increase in roughness. Moreover, they also proposed an extended force balance CHF model to incorporate the effect of surface roughness.

Although surface modification techniques have shown promising results but at CHF, the performance of these boiling surfaces is hampered by the formation of vapor blanket which acts as an insulator which results in abrupt increase in the surface temperature causing a catastrophic system burnout. Thus, it is imperative to estimate the CHF for any surface-liquid combination under various operational conditions beforehand. So far, most of the correlations and semi-analytical models, to predict the CHF, reported in the literature are based on five popular CHF triggering mechanisms i.e., hydrodynamics instability, hot/dry spot, bubble interface, and interfacial lift-off model [1,11].

Mudawar et al. [21] studied Water and FC-72 to understand the CHF triggering mechanism for the pool boiling on vertical surfaces. Through wetting front visualization during the boiling of FC-72, they found that mechanism for vertical surfaces near CHF is fairly similar to the flow boiling. They concluded that CHF occurs when the momentum of vapor becomes higher than the pressure force of interfacial curvature of wetting front and consequently lift-off the interface from the wall. Based on this, they developed an interfacial lift-off model. This model was originally developed for vertical surfaces, however, through data available in literature, Howard and Mudawar found that orientation effects on CHF for $0 < \theta < 90^\circ$ are very weak, which indicates that this model is also valid for $\theta = 0^\circ$ [22]. Bailey [23] tested methanol, water, and Pentane and proposed a correlation for CHF at ambient conditions. Guan et al. [24] tested Pentane, Hexane, and FC-72 under saturated conditions at three chamber pressures (150, 300, and 450 kPa) during the pool boiling on horizontal surfaces. They stated that CHF occurs when liquid microlayer is removed due to the vapor momentum flux and proposed a model similar to the lift-off model of Zuber [25]. After analyzing the most commonly used CHF correlations, Liang and Mudawar [11] recommended that to predict CHF for horizontal upward facing surfaces, Lienhard and Dhiri hydrodynamic model [26] and Mudawar et al. interfacial lift-off model [21] can be preferably used.



(a)



(b)

Fig. 2. Selection of input parameters for CHF prediction and (b) hyper-parameters tuning of DNN [41].

Table 2

Range of the output parameters.

Parameters	Range
Substrate material thermal conductivity	$1.14 \leq k_{mat} \leq 385 \text{ W/m-K}$
Operational pressure	$0.025 \leq p \leq 0.4 \text{ MPa}$
Surface Inclination	$0 \leq \theta \leq 180^\circ$
Wall superheat	$0.5 \leq \Delta T \leq 95 \text{ K}$
Saturation Temperature	$-196 \leq T \leq 100 \text{ }^\circ\text{C}$
Roughness	$0.02 \leq Ra \leq 10.3 \text{ } \mu\text{m}$
Heat flux	$0.1 \leq q \leq 2161 \text{ KW/m}^2$

Moreover, in various applications, the heat source can be at an angle other than 0° (horizontal position). For instance, a CPU can be mounted at various angles other than horizontal or in pressurized water reactors in which molten mixture is relocated to the bottom head of the vessel or in various space applications where system is continuously rotating so, it is also necessary to predict the impact of orientation on the CHF value [12,27,28]. Vishnev [29] was the first to correlate the effect of orientation on CHF. He developed a correlation based on the boiling data of liquid Helium and reported that if the boiling surface has a diameter of 10–15 mm, then surface dimensions will not have much impact on the CHF. El-Genk and Bostanci [30] experimentally studied the effect of inclination on the CHF during the pool boiling of HFE-7100 at atmospheric pressure. They noted that CHF decreases slightly for the inclination angle ($\theta \leq 90^\circ$) and decreases drastically for the inclination angle ($\theta \geq 90^\circ$) due to formation of many bubbles around the heating surface. They also developed a correlation incorporating the inclination effect which predicted the CHF within $\pm 3\%$ of experimentally measured CHF. Arik and Bar-Cohen [31] investigated the pool boiling of dielectric liquids and developed a correlation for critical heat flux based on the hydrodynamic instability theory of Zuber [25]. They noted that CHF values of HFE-7100, HFE-7200 were almost 50 % higher than that of FC-72. Brusstar and Merte [32,33] examined the pool boiling of R-113 and developed a correlation by taking into account bubbles sliding effect which is valid for the inclination angle between 90 and 180° . They concluded that buoyancy force has a significant impact on the CHF. However, for velocities higher than 0.55 ms^{-1} effect of orientation on the CHF was minimal. Lienhard and Dhir [26,34] re-evaluated the hydrodynamic theory of Zuber and proposed the correlation to predict the CHF by incorporating the geometrical effect and modified the vapor velocity condition. Further details can be found in recent review articles

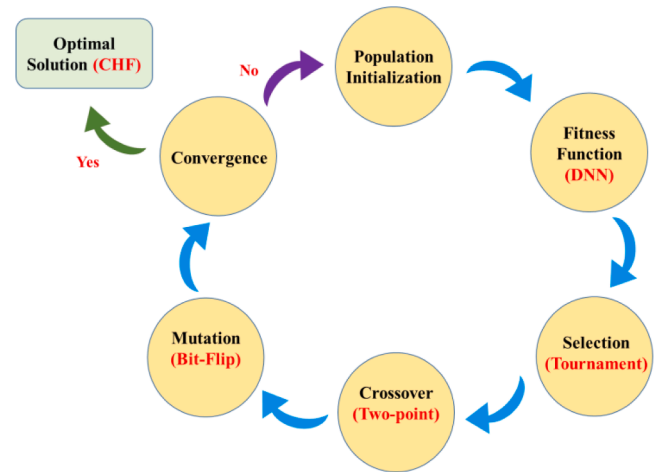
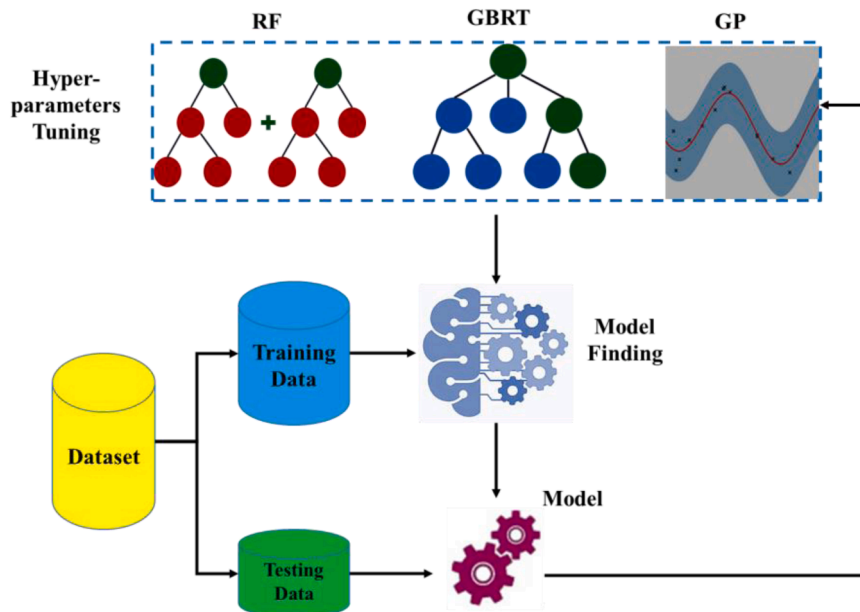
by Li et al. [35] and Wang et al. [36].

As far as the predictions about the CHF are concerned, various CHF semi empirical and empirical correlations have been reported by many investigators [37,38]. However, MAE of these correlations is as high as 163 % [11]. This large error is due to the complex nature of the boiling phenomenon which cannot be completely captured by empirical correlations. On the other hand, machine learning (ML) model performs best

Table 3

Optimal hyperparameters.

Parameter	Optimal Value
No of dense layers	4
No of dense nodes	25
Activation function	ReLU
Learning rate	0.0067
Decay rate	0.0029
Optimizer	Adam
Epochs	89
Kernel initializer	He-uniform
Batch size	47

**Fig. 4.** Working of the Genetic Algorithm.**Fig. 3.** K-fold search using Bayesian surrogate models.

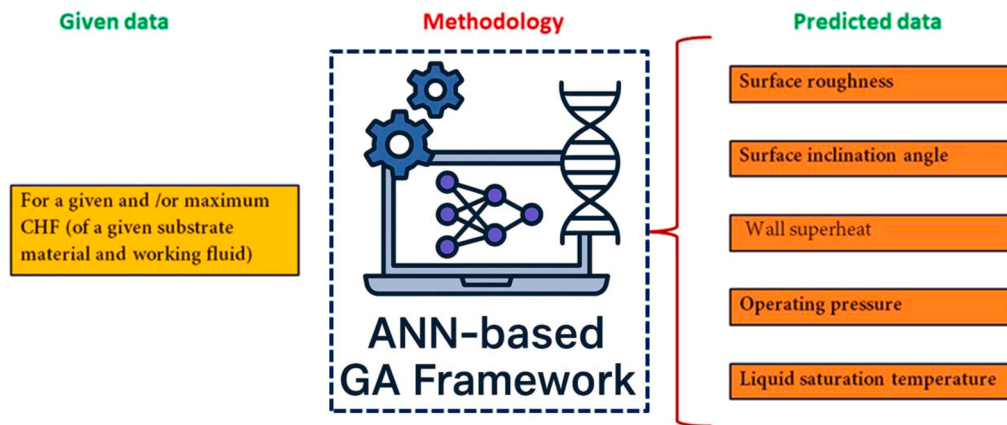


Fig. 5. An overview of the methodology of surface morphological and operating parameters optimization for the desired CHF value.

when given as many details as possible because ML models can spread the data into high dimensions and can understand the underlying convoluted relationship better than the empirical correlations with much lower MAE [39]. This problem can be solved by using advanced machine learning techniques. ML models have already been employed to predict various parameters such as HTC, CHF, and pressure drop in the two-phase thermal systems [40–44]. Also, bubble departure diameter has been predicted with a decent accuracy through DNN models [45]. Also, in our previous study, we employed ML to predict CHF by providing 05 key parameters (which are surface roughness, surface inclination, wall superheat, saturation temperature, and operating pressure) as an input to the deep neural network model (DNN) [41].

This study introduces an innovative inverse modeling framework that combines DNN with a GA model to determine optimal surface morphological and operational parameters for achieving target CHF values in pool boiling systems. Unlike existing empirical correlations, which demonstrate limited accuracy and generalizability, the proposed DNN model, trained on an extensive dataset comprising 13,000 data points and 29 fluids, achieves significantly enhanced prediction accuracy ($MAE = 48$, $R^2 = 0.97$). The GA component addresses the inverse design challenge, facilitating tailored parameter selection for various working fluids and substrate properties. Comparative results exhibit strong alignment with experimental observations, thereby validating both performance trends and parameter sensitivities reported in previous studies. This integrated approach provides a robust predictive tool for advancing CHF management in thermal systems.

In this study, GA was chosen due to its inverse optimization problem solving capabilities and inherent flexibility without requiring gradient or explicit symbolic relationships. Although Physics-Informed Neural Networks (PINNs) and sparse regression techniques such as Ridge/Lasso have been used by numerous researchers, these methods generally lack the direct optimization capability and global search efficiency offered by GA. So, GA is a superior framework and meets the objectives of this study. The manuscript is structured as follows: Section 2 discusses methodology which includes data collection, pre-processing, development of deep neural network (DNN), and optimization procedure using Genetic Algorithm (GA). Section 3 presents the results and discussion, followed by performance analysis of DNN and concludes with the predictions of GA model. Section 4 summarizes the key findings learned from this study.

2. Methodology

2.1. Data Collection and Pre-processing

A dataset of 13,000 points were collected from 42 pool boiling

studies incorporated with 29 working fluids on various rough and pored surfaces under wide range of operational conditions. Details about the dataset can be found in our previous study [41]. Brief details about the dataset are given below in Table 1.

An overview of the methodology to tailor the optimal structural and operational parameters for a desired CHF value for a range of surface-liquid combinations is given in Fig. 1. The steps include collection of pool boiling CHF data, data preprocessing, hyper-parameters tuning, physics and correlations based selection of the input parameters, training and testing of the deep neural network model, interpretation of the DNN predictions by using SHAP library (explainable artificial intelligence) to highlight the contributions of various input parameters, and then development of genetic algorithm for the optimization task.

2.2. Selection of input parameters

To identify the key parameters, we first explored the CHF correlations reported in the literature and selected the commonly used parameters in various correlations. Later on, in order to find the relationship between these parameters and the CHF database, we also employed three statistical matrices i.e., Spearman, Pearson, and Kendal correlation to identify the degree of the relationship. After analyzing through both methods, we picked the top six parameters. Fig. 2(a) shows the selection of input parameters. Further details can be found in a study by Sajjad et al. [41].

We can see that surface inclination angle, surface roughness, operating pressure, wall-superheat, saturation temperature, and thermal conductivity of the substrate are the most influential parameters in predicting the CHF using DNN model for the considered data. Thus, extending that we developed the GA model, which predicts these optimal parameters when desired CHF value is given as an input. The model was trained on a large database and thus can predict the optimal parameters in the range of parameters given below in Table 2.

2.3. Deep neural network model (DNN) development

A neural architecture search is performed using Bayesian surrogacy with Gaussian process, random forest, and gradient boost regression trees. The learning rate, Adam decay, input nodes, dense layers, dense nodes, batch size, and activation function are among the investigated hyper-parameters. To obtain a robust model, these parameters/hyper-parameters must be tuned. To do so, we employed Bayesian optimization which gave us an optimal combination of hyperparameters as shown in Fig. 2(b). The optimal parameters are highlighted with star symbols. While choosing hyperparameters to assess the actual fitness function, Bayesian surrogate models provide a probabilistic model of

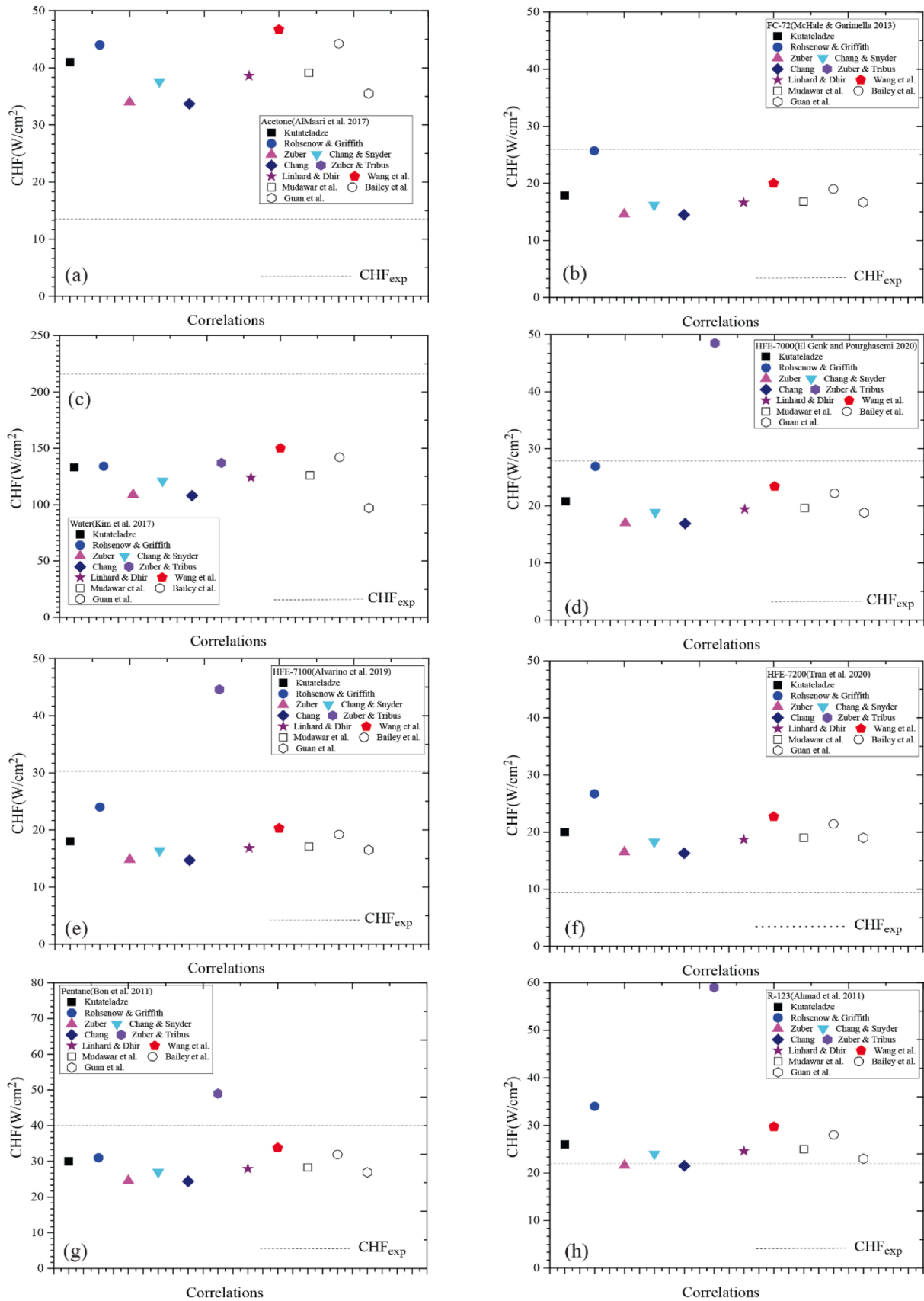


Fig. 6. Predicted CHF (W/cm²) by empirical correlations for (a) Acetone, (b) FC-72, (c) Water, (d) HFE-7000, (e) HFE-7100, (f) HFE-7200, (g) Pentane, and (h) R-123 [41].

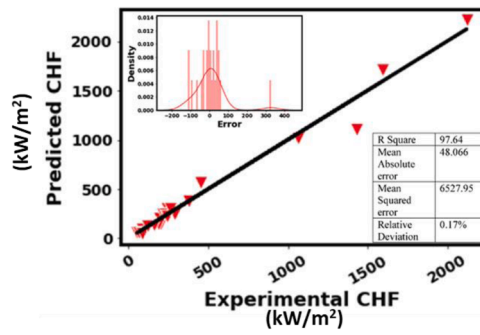


Fig. 7. Predicted CHF (kW/m^2) using optimized DNN test dataset.

fitness function. Wherever the acquisition function (selection function) is maximized, the corresponding hyperparameters are selected. The k-fold search (see Fig. 3) revealed the parameters to build the optimal deep neural network (see Table 3). For more information about the DNN model, please refer to Sajjad et al. [41].

2.4. Genetic Algorithm optimization

Genetic algorithm (GA) is a stochastic search algorithm inspired by basic biological principles of biological evolution where only the fittest living organisms dominate the weaker ones. It mimics evolution mechanisms like selection, crossover, and mutation. At first a population with individuals of varying fitness is initiated in which each individual is made up of genes, features, and several characters. Later, the fitness of each individual is evaluated, and the fittest ones produce new offspring,

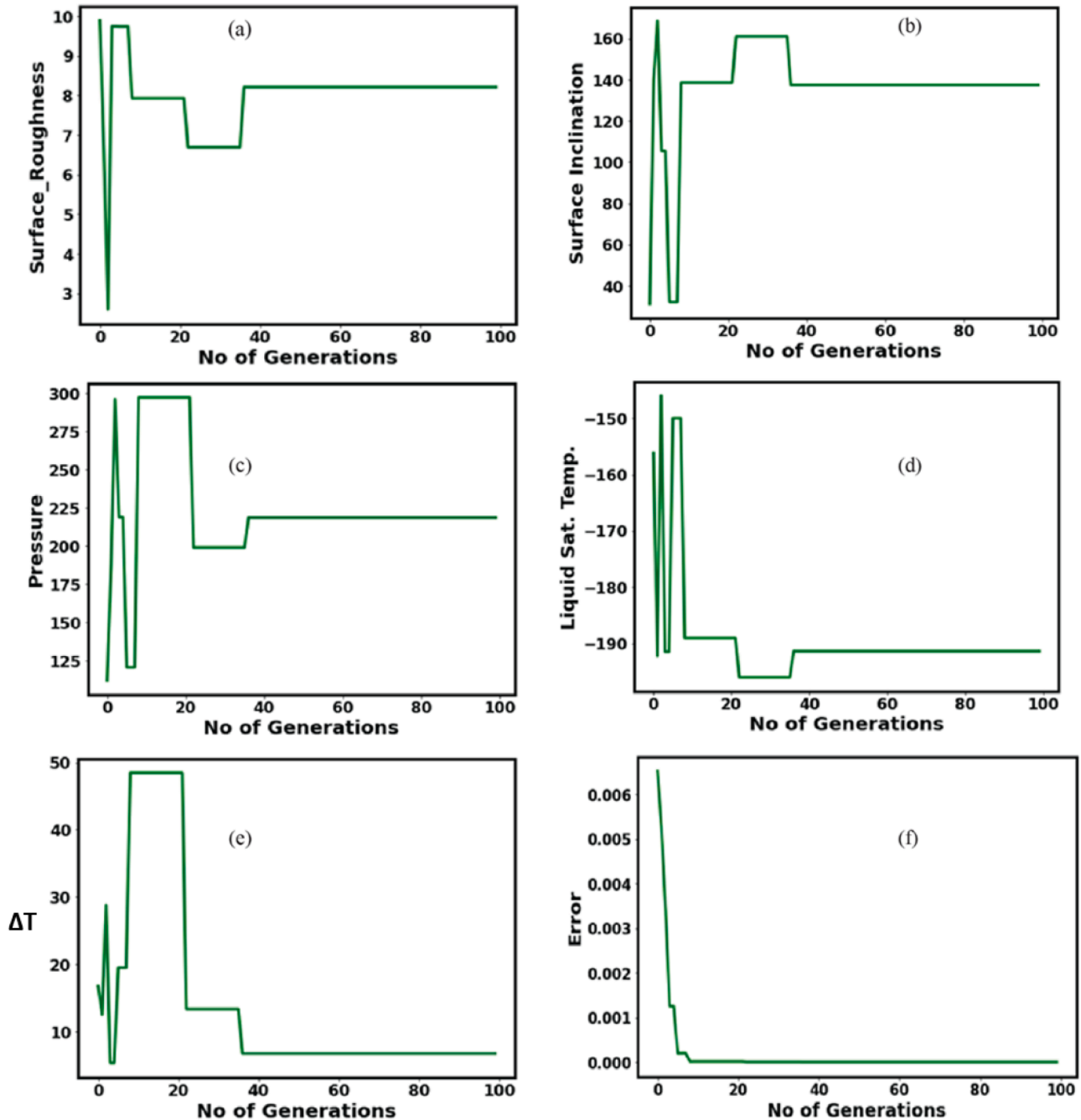


Fig. 8. Suggested values of optimal parameters(a-e) and error plot (f) at a desired CHF value of 120 kW/m^2 .

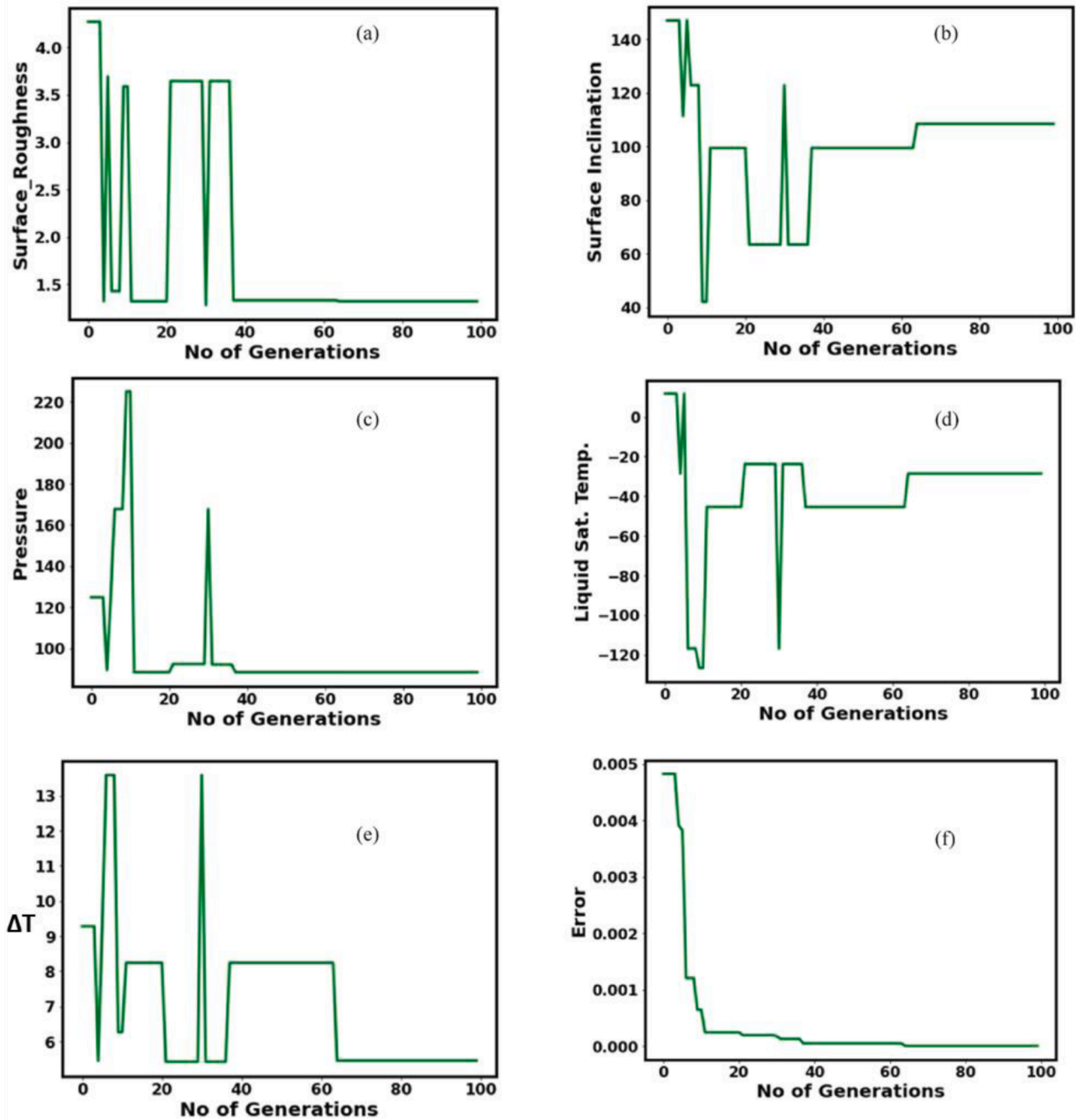


Fig. 9. Suggested values of optimal parameters(a-e) and error plot (f) at a desired CHF value of 252 kW/m^2 . Suggested optimal values of the parameters are also shown in Figs. 10 and 11.

Table 4

(a) Range of the input parameters and (b) suggested optimal parameters.

(a)		
Input Parameters	Minimum	Maximum
Log ΔT	1.69	3.927
Surface Roughness	0.039	10.3
Surface Inclination	0	180
Pressure	85	300
Liquid Sat. Temp.	-196.15	100
(b)		
Parameters	Optimal solutions	
Surface Roughness	5.626	9.038
Surface Inclination	72.48	168.81
Pressure	156.89	229.61
Liquid Sat. Temp.	-105.5	-190.9
Log ΔT	2.149	2.193
CHF	252.11	119.26

and previous information is passed to the offspring. Later, mutation is also performed with parents of random characteristics and previous ones are discarded. The process is repeated till the optimal solution is obtained or the maximum number of iterations has reached. A schematic about the working of the GA algorithm is shown in Figs. 4 and 5 shows the methodology for designing an optimal surface for a given CHF value as an input. We can see that when GA is provided with a target CHF value, it predicts the key design and operating parameters such as surface roughness, inclination angle, heat flux, operational pressure and

liquid saturation temperature.

3. Results and discussion

3.1. CHF Predicted by existing empirical correlations

The existing empirical correlations for CHF were evaluated on the consolidated database. Fig. 6 shows the predictions by various commonly used CHF correlations on the experimental data. We can see that most of the correlations performs poorly when tested on wide range of operating conditions [41].

3.2. CHF predicted by deep neural network

To examine the accuracy of the proposed deep neural network we used 30 % of the test data (not used for training the model) to predict the CHF values and compared it with the experimentally reported data as shown in Fig. 7. As shown in Fig. 7, model was able to precisely predict the CHF values with an MAE of 48 and R^2 score of 0.97.

3.3. Prediction of optimal operational conditions using GA

To predict the optimal parameters at the desired value of critical heat flux, we tested commonly used working fluids like HFE-7000, Water, HFE-7200, and FC-72 at typical CHF values for the substrate having thermal conductivity between 205–385 W/m-K. Figs. 8 and 9 show the convergence of GA models, with the increase in number of generations,

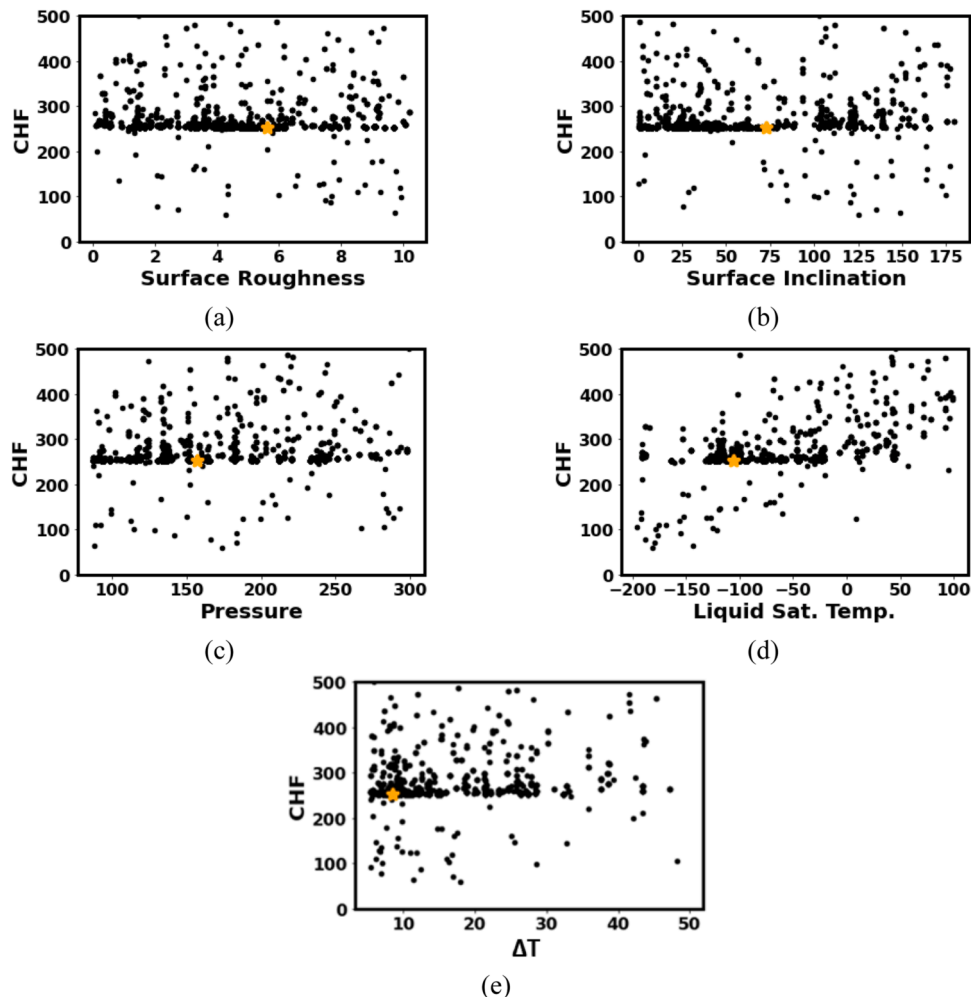


Fig. 10. Suggested values of optimal parameters (a-e) at a desired CHF value of 120 kW/m².

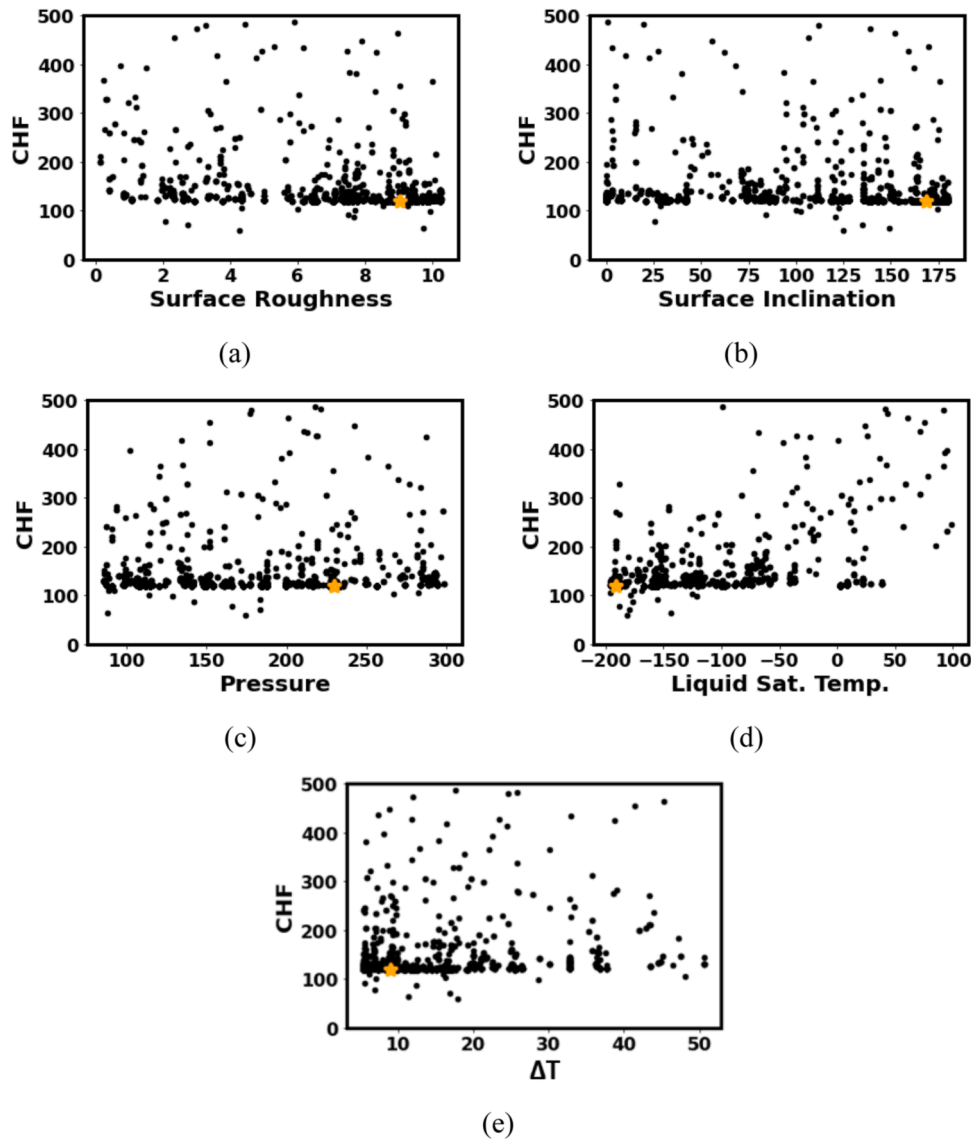


Fig. 11. Suggested values of optimal parameters(a-e) at a desired CHF value of 252 kW/m².

Table 5

(a) Range of the input parameters and (b) suggested optimal parameters.

(a)		
Input Parameters	Minimum	Maximum
Log ΔT	0.42	3.76
Surface Roughness	0.035	7.8
Surface Inclination	0	180
Pressure	101	115
Liquid Sat. Temp.	51	100
(b)		
Parameter	Optimal solutions	
Surface Roughness	7.106	5.919
Surface Inclination	33.216	1.425
Pressure	110.820	109.426
Liquid Sat. Temp.	99.793	98.418
Log ΔT	3.0972	2.932
CHF	848.988	1595.95

during the training process. Initially model shows fluctuations in the values of all parameters, which is understandable as the model is trying to learn but after 40 iterations model converges quickly and gets stable afterward.

Table 4 shows the search space and suggested optimal values of input parameters for the HFE-7000 at a given CHF on the substrate having thermal conductivity of 385 W/m-K. From Table 4(b) we can see the with the increase in the required CHF value, values of inclination angle decreases because at higher inclination it becomes difficult to remove the vapor which causes CHF to occur earlier [1]. Model also suggested that to obtain a higher heat transfer performance saturation temperature needs to be high which has been reported in many studies for instance Mehdi et al. [6] found that heat transfer increased with the increase in saturation temperature due to favorable bubble dynamics. Similarly model also shows the decrease in surface roughness, wall-superheat, and system pressure. Suggested optimal values of the parameters are also shown in Figs. 10 and 11.

Table 5 shows the range of input parameters and predicted values of input parameters for the CHF of 848.98 kW/m² and 1595.95 kW/m² during the pool boiling of water on the substrate having thermal conductivity of 205 W/m-K. We can see that with the increase in desired CHF value there is a slight decrease in all parameters except surface inclination which seems understandable for instance to have higher CHF value, wall-superheat should decrease. Also, maximum enhancement in CHF has been reported during the pool boiling when the boiling surface was in horizontal position [12]. The optimization process to obtain

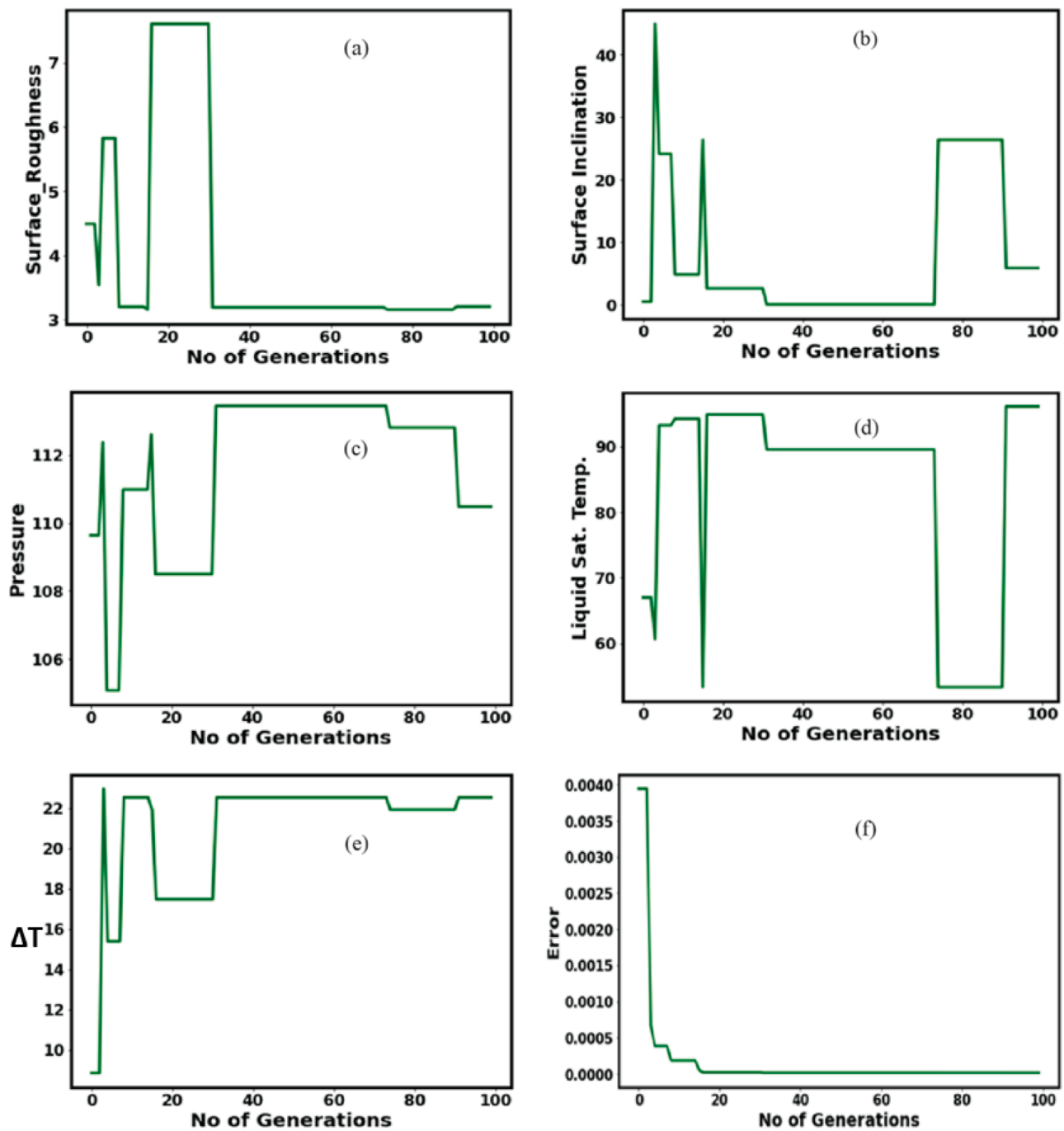


Fig. 12. Suggested values of optimal parameters(a-e) and error plot(f) at a desired CHF value of 848.98 KW/m².

optimal parameters have been shown in Figs. 12 and 13. At optimal values we can see that model shows a flat line for each parameter while at the start there are many spikes in the training curves. The suggested optimal values are also shown in Figs. 14 and 15.

Table 6 shows the range of input parameters and predicted values of input parameters for the CHF of 41.11 kW/m² and 93.8 kW/m² during the pool boiling of HFE-7200 on the substrate having thermal conductivity of 205 W/m-K. Other than ΔT and surface roughness all other parameters' values were increased with the increased required CHF value. This trend is similar to the HFE-7000 probably because both liquids have similar properties i.e., of a highly wetting nature. We can see that with the two-time increase in CHF value an almost two-fold increase in saturation temperature is suggested by the model along with a 42 % decrease in the wall-superheat (from 65 to 38). Moreover,

models also suggest almost the same inclination angle of the boiling surface. Training pattern plots (not shown here for compactness) followed the same trend i.e., there is a spike till some generations (40 in this case) and then after that it shows a constant value for each parameter by indicating a flat line. The suggested optimal values are also shown in Figs. 16 and 17.

In the last case, we predicted the optimal parameters for the FC-72 at a given CHF value. The model converges within 50 generations (iterations) and similar to the earlier cases, once it achieved an optimal value it shows a flat line which is the indication of the optimal point. For the conciseness, the training plots are not shown here. Table 7 shows the range of input parameters and predicted values of input parameters for the CHF of 164.172 kW/m² and 260.53 kW/m² during the pool boiling of FC-72 on the substrate having thermal conductivity of 205 W/m-K.

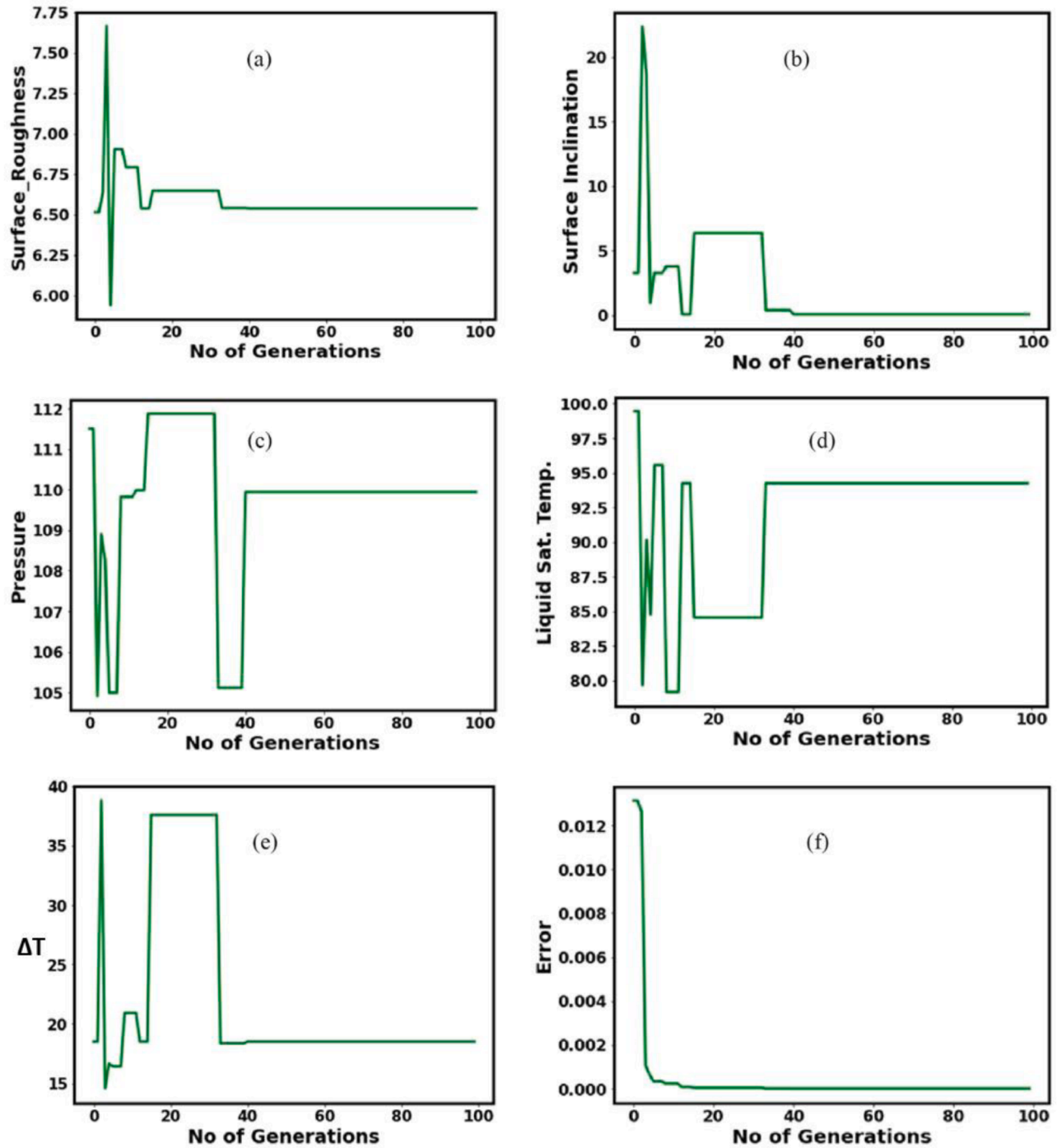


Fig. 13. Suggested values of optimal parameters(a-e) and error plot(f) at a desired CHF value of 1595.95 kW/m^2 .

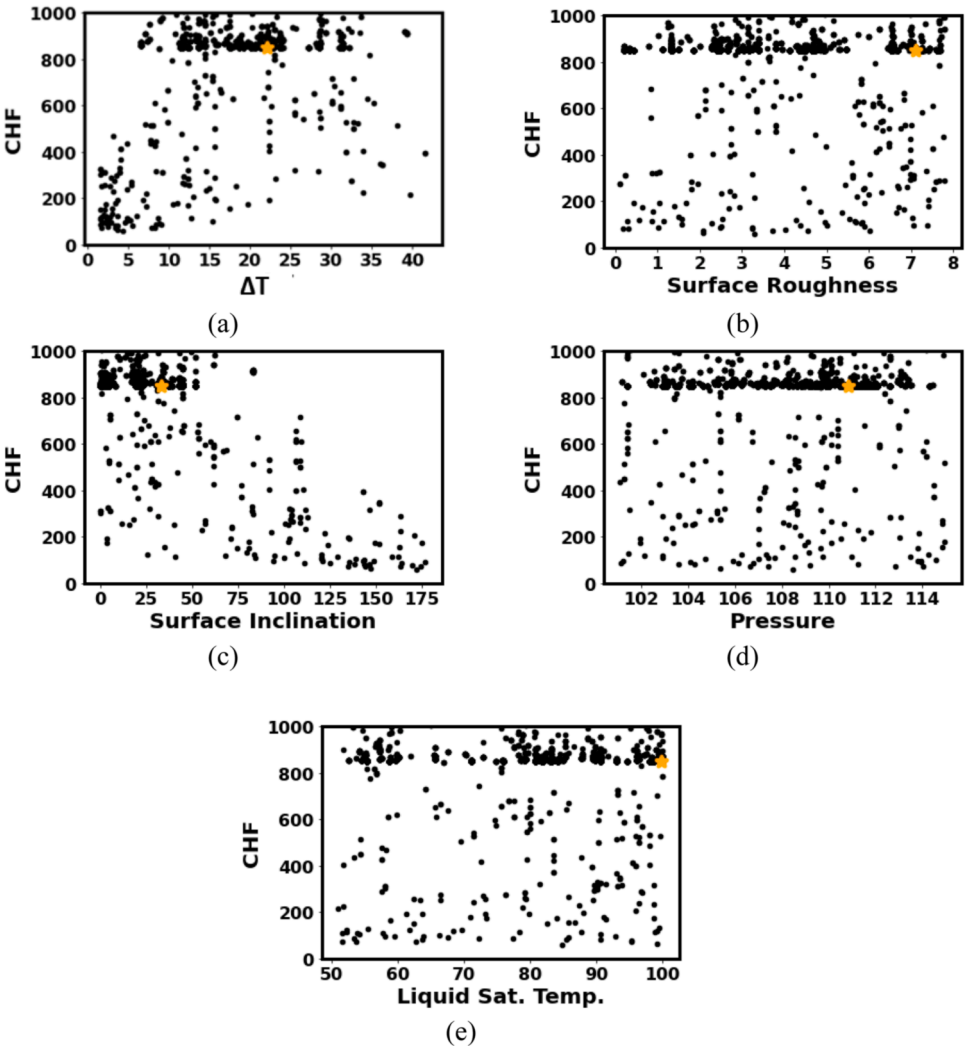


Fig. 14. Suggested values of optimal parameters (a-e) at a desired CHF value of 848.98 kW/m².

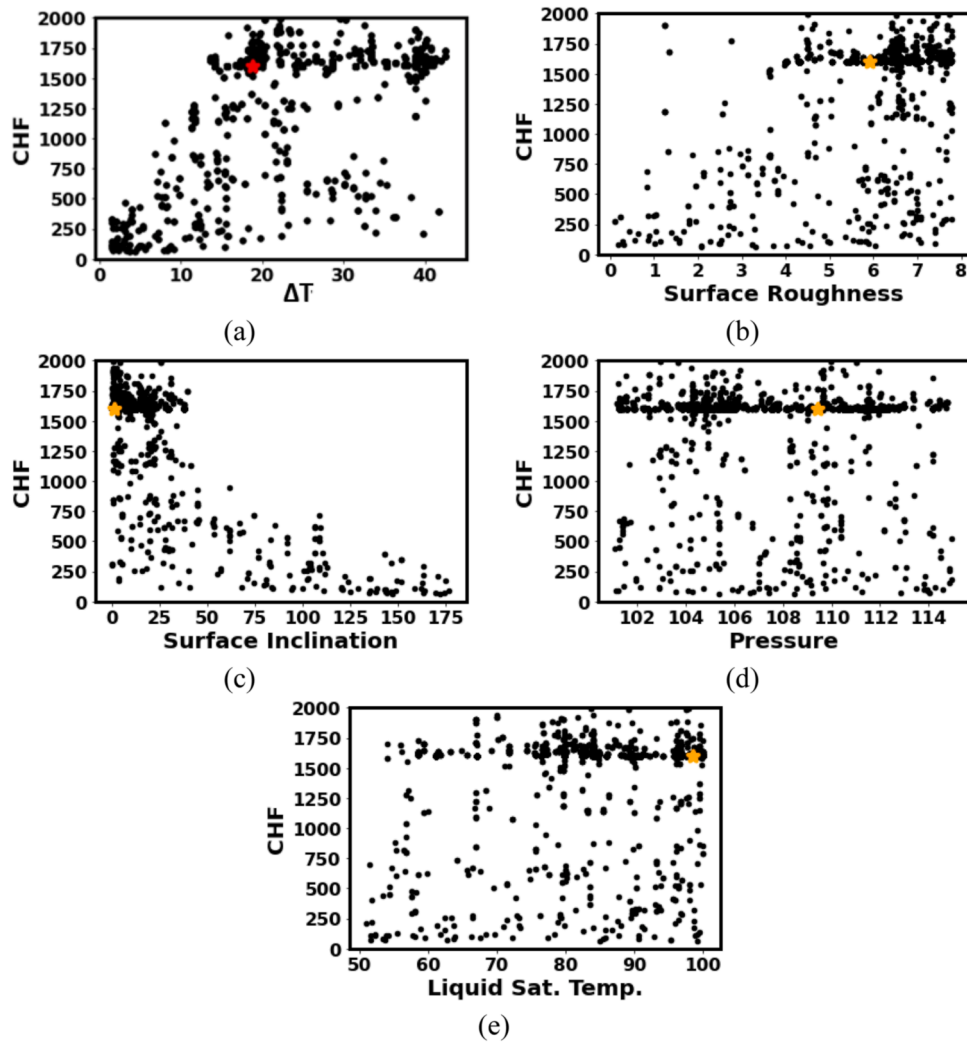


Fig. 15. Suggested values of optimal parameters (a-e) at a desired CHF value of 1595.95 kW/m^2 .

Table 6

(a) Range of the input parameters and (b) suggested optimal parameters.

(a)		
Input Parameters	Minimum	Maximum
Log ΔT	2.84	4.26
Surface Roughness	0.032	1.22
Surface Inclination	0	180
Pressure	101	400
Liquid Sat. Temp.	32	100
(b)		
Parameter	Optimal solutions	
Surface Roughness	0.0035	0.0039
Surface Inclination	170	167
Pressure	241	102
Liquid Sat. Temp.	86.7	54.24
Log ΔT	3.673	3.94
CHF	93.8	41.11

From the Table 7 we can see that for a CHF value of 164.172 kW/m^2 model predicts saturation temperature of 78.9°C and surface inclination to be 108° for the surface having roughness of $1.2 \mu\text{m}$. Moreover, with the 40 % increase in CHF value of 260.53 kW/m^2 , model predicts 70 % reduction in surface inclination. Suggested optimal values are also shown in Figs. 18 and 19.

4. Conclusion

In this study, we extended the DNN findings and proposed a GA based optimization model to suggest the optimal input design/operational parameters at a desired CHF value. The data was collected from 42 studies which include pool boiling of 29 working fluids on various rough and pored surfaces under wide range of operational conditions.

The following key conclusions were drawn:

- The existing empirical correlations did not adequately predict the CHF for different surface-liquid combinations.
- The CHF prediction accuracy was significantly improved when predicted with the help of DNN. The DNN model predicts with an MAE of 48 and R^2 score of 0.97 which is a sure improvement compared to the empirical correlations.
- GA models can be implemented to obtain optimal design and operational parameters for the desired value of CHF.
- To determine the optimal parameters for the necessary value of critical heat flux, we examined commonly used working fluids such as HFE-7000, Water, HFE-7200, and FC-72 at typical CHF values for the substrate having thermal conductivity between 205 and 385 W/m-K .
- For the HFE-7000 at a given CHF (120 kW/m^2 , 252 kW/m^2) on the substrate having thermal conductivity of 385 W/m-K , we can see that with the increase in the required CHF value, values of inclination

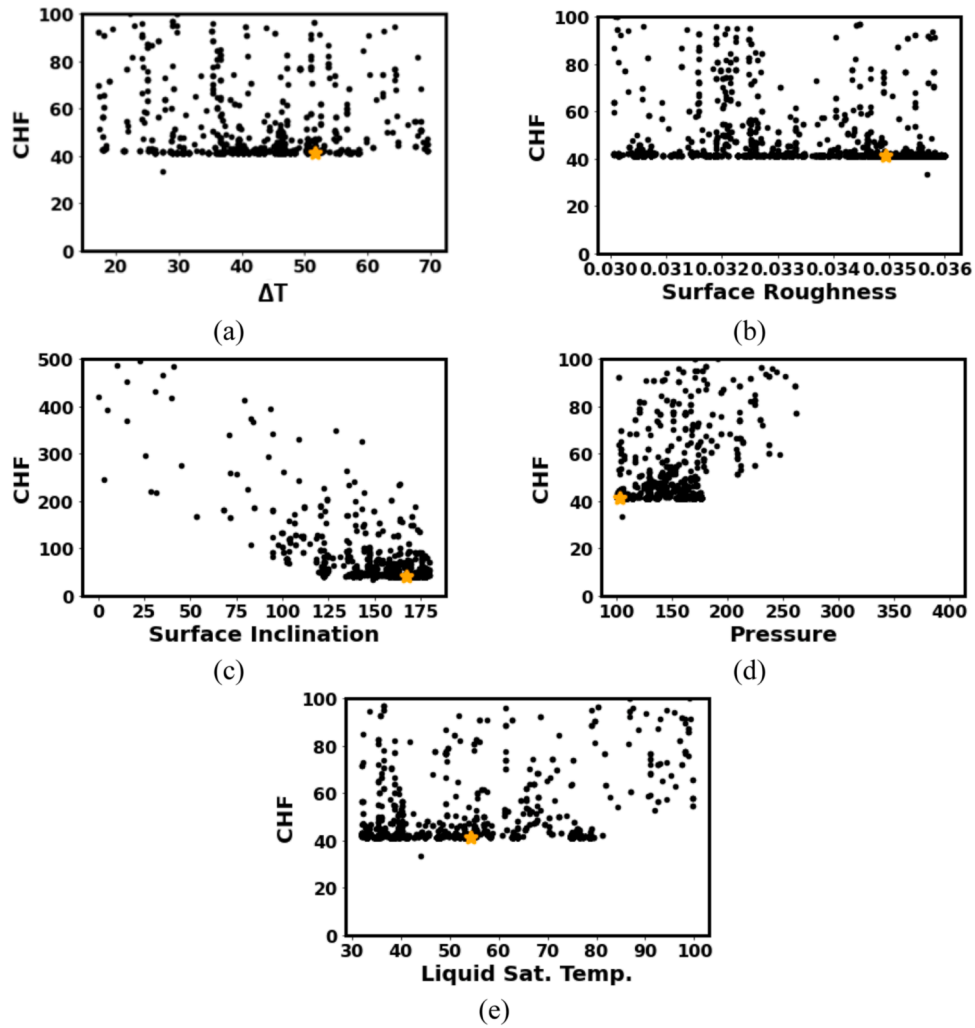


Fig. 16. Suggested values of optimal parameters (a-e) at a desired CHF value of 41.11 kW/m².

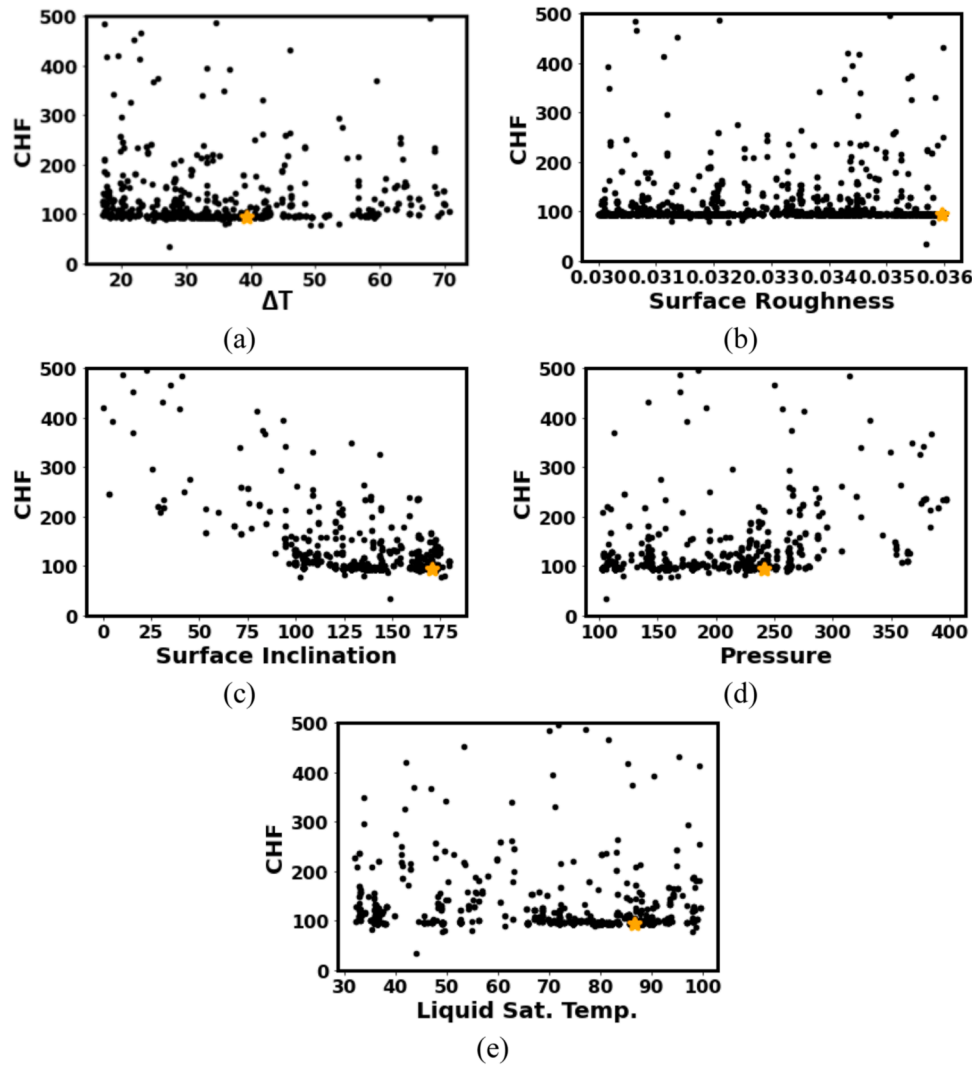


Fig. 17. Suggested values of optimal parameters (a-e) at a desired CHF value of 93.8 kW/m².

Table 7

(a) Range of the input parameters and (b) suggested optimal parameters.

(a)		
Input Parameters	Minimum	Maximum
Log ΔT	0.3	3.25
Surface Roughness	0.03	1.22
Surface Inclination	0	180
Pressure	60	101
Liquid Sat. Temp.	56	100
(b)		
Parameters	Optimal solutions	
Surface Roughness	1.2	0.928
Surface Inclination	108	23.15
Pressure	75.5	68.28
Liquid Sat. Temp.	78.9	58.59
Log ΔT	2.19	3.02
CHF	164.172	260.53

- angle decrease because at higher inclination it becomes difficult to remove the vapor which causes CHF to occur earlier.
- For HFE-7200 at 41.11 kW/m² and 93.8 kW/m² on the substrate having thermal conductivity of 205 W/m-K, other than ΔT and

surface roughness, all other parameters' values were increased with the increase in the required CHF value. This trend is similar to the HFE-7000 probably because both liquids have similar thermophysical properties i.e., of a highly wetting nature.

- Similarly, For FC-72 at CHF of 164.172 kW/m² and 260.53 kW/m² on the substrate having thermal conductivity of 205 W/m-K, we can see that for a CHF value of 164.172 kW/m², model predicts saturation temperature of 78.9 °C and surface inclination to be 108° for the surface having roughness of 1.2 μ m.
- The GA-driven inverse design framework exhibited physically meaningful behavior. For example, with increasing target CHF, the model consistently recommended reduced inclination angles, which correspond with known limitations in vapor removal at steep angles. Similarly, trends in surface roughness and saturation temperature were consistent with boiling enhancement mechanisms reported in prior experimental studies. These observations confirm that, despite being purely data-driven, the model captures fundamental thermo-fluidic phenomena, reinforcing its applicability for predictive design in thermal systems.
- The proposed models can be used as a predictive/design tool to estimate CHF and optimal surface and operating parameters to obtain a desired CHF value for the thermal systems.

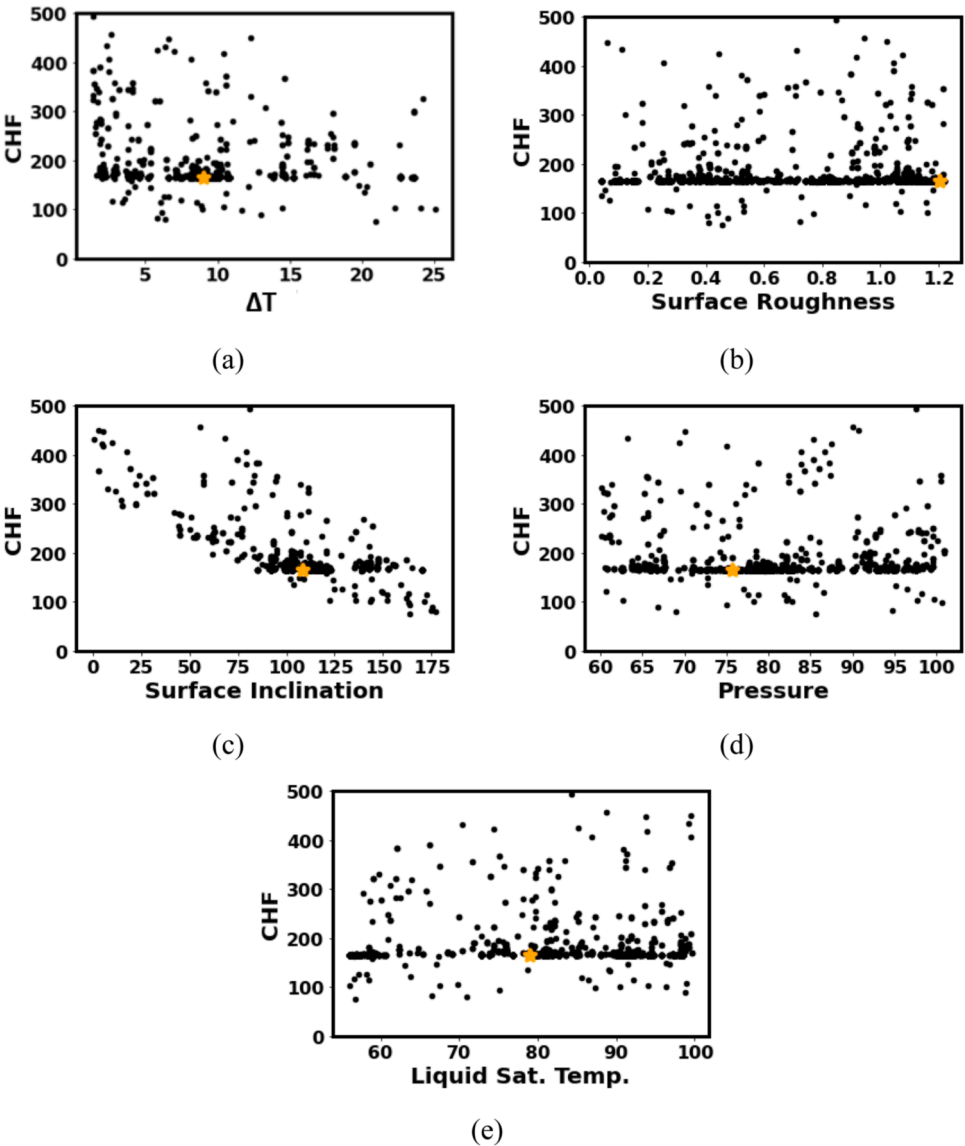


Fig. 18. Suggested values of optimal parameters (a-e) at a desired CHF value of 164.17 kW/m².

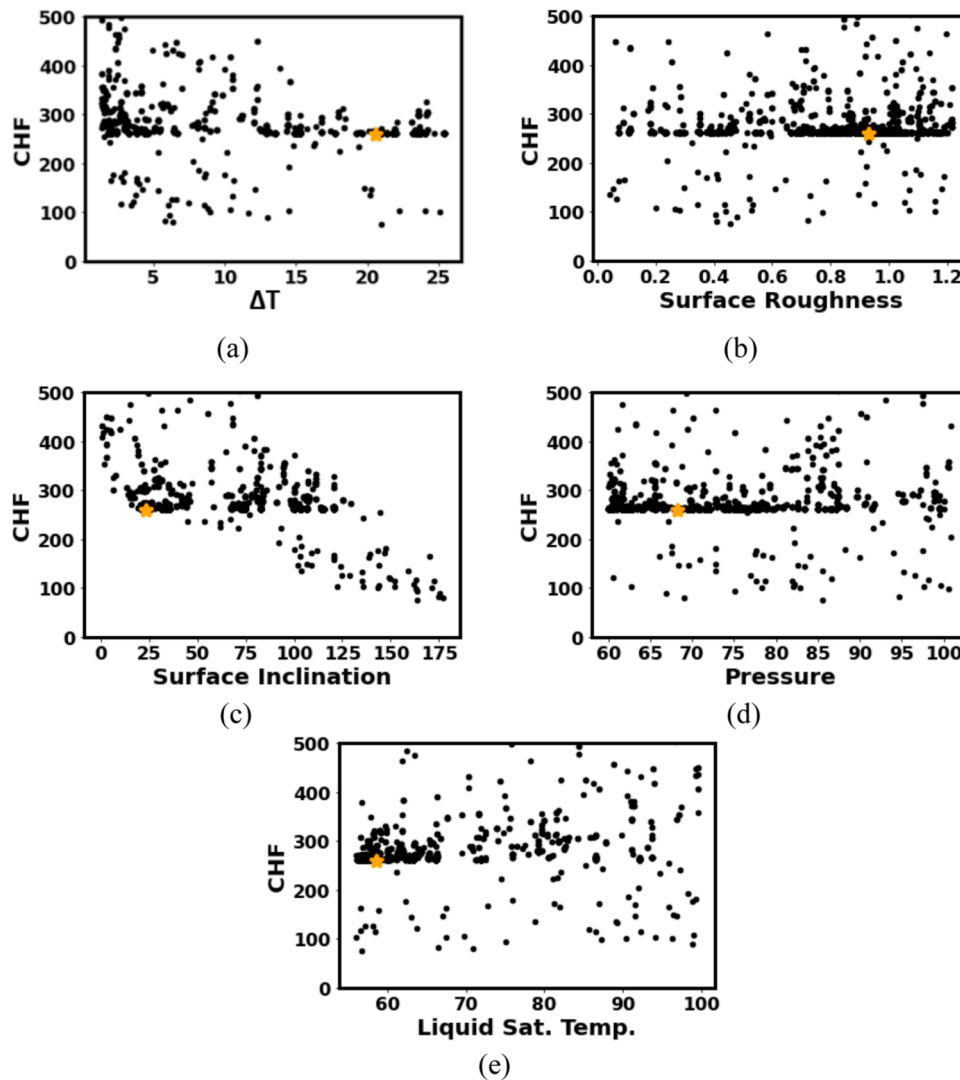


Fig. 19. Suggested values of optimal parameters (a-e) at a desired CHF value of 260.53 kW/m².

CRedit authorship contribution statement

Uzair Sajjad: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Data curation, Conceptualization. **Sadaf Mehdi:** Writing – original draft, Writing – review & editing, Conceptualization, Visualization. **Syed Shoaib Hassan Zaidi:** Visualization, Software, Validation, Resources. **Hafiz Muhammad Ali:** Writing – review & editing, Supervision, Formal analysis, Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

References

- [1] G. Liang, I. Mudawar, Pool boiling critical heat flux (CHF)–Part 1: review of mechanisms, models, and correlations, *Int. J. Heat. Mass Transf.* 117 (2018) 1352–1367.
- [2] I. Mudawar, Assessment of high-heat-flux thermal management schemes, *IEEE Trans. Compon. Packag. Technol.* 24 (2) (2001) 122–141.
- [3] S. Mehdi, S. Nannapaneni, G. Hwang, Structural-material-operational performance relationship for pool boiling on enhanced surfaces using deep neural network model, *Int. J. Heat. Mass Transf.* 198 (2022) 123395.
- [4] F. Ahmad, S. Kim, M. Meyer, J. Hartwig, I. Mudawar, Saturated nucleate pool boiling of cryogenic fluids: review of databases, assessment of existing models and correlations, and development of new universal correlation, *Int. J. Heat. Mass Transf.* 231 (2024) 125807.
- [5] F. Ahmad, M. Meyer, J. Hartwig, I. Mudawar, Saturated pool film boiling of cryogenic fluids: review of databases, assessment of existing models and correlations, and development of new universal correlation, *Int. J. Heat. Mass Transf.* 235 (2024) 126190.
- [6] S. Mehdi, Y. Shah, A. Mondal, N.-H. Kim, Bubble dynamics of R-134a boiling in enhanced surfaces having pores on sub-tunnels, *Int. J. Heat. Mass Transf.* (2020) 155.
- [7] S. Mehdi, N.-H. Kim, Nucleate pool boiling of R-123 on pored surfaces, *J. Enhanc. Heat Transf.* 27 (7) (2020).
- [8] S. Mehdi, Y. Shah, N.-H. Kim, Bubble dynamics of R-123 and R-134a on pore/sub-tunnel surfaces, *Int. J. Therm. Sci.* 177 (2022) 107543.
- [9] M. Borumand, G. Hwang, Enhanced pool boiling critical heat flux on tilted heating surfaces using columnar-post wicks, in: *ASME International Mechanical Engineering Congress and Exposition*, American Society of Mechanical Engineers, 2021 pp. V011T011A028.
- [10] M. Borumand, G. Hwang, Enhanced pool boiling critical heat flux on tilted heating surfaces using columnar-post wicks (IMECE2021-70054), in: *Proceedings of the ASME 2021 International Mechanical Engineering Congress and Exposition (IMECE)*, 2021.
- [11] G. Liang, I. Mudawar, Pool boiling critical heat flux (CHF)–Part 2: assessment of mechanisms and correlations, *Int. J. Heat Mass Transf.* 117 (2018) 1368–1383.
- [12] M. Egbo, M. Borumand, Y. Nasersharifi, G. Hwang, Surface orientation effects on pool-boiling with plain and enhanced surfaces, *Appl. Therm. Eng.* (2021) 117927.

- [13] S. Mehdi, A. Mondal, N.-H. Kim, Pool boiling heat transfer of R-123 on pored surfaces, in: Society of Air-conditioning and Refrigerating Engineers of Korea (SAREK), 2019, pp. 610–613.
- [14] A. Mondal, S. Mehdi, N.-H. Kim, Pool boiling heat transfer of R-134a on pored surfaces, in: Society of Air-conditioning and Refrigerating Engineers of Korea (SAREK), 2018, pp. 453–455.
- [15] G. Liang, I. Mudawar, Review of nanoscale boiling enhancement techniques and proposed systematic testing strategy to ensure cooling reliability and repeatability, *Appl. Therm. Eng.* (2021) 184.
- [16] H. Kim, M. Kim, Experimental study of the characteristics and mechanism of pool boiling CHF enhancement using nanofluids, *Heat Mass Transf.* 45 (7) (2009) 991–998.
- [17] J. Chang, S. You, Boiling heat transfer phenomena from microporous and porous surfaces in saturated FC-72, *Int. J. Heat Mass Transf.* 40 (18) (1997) 4437–4447.
- [18] G.-S. Hwang, M. Kaviani, Critical heat flux in thin, uniform particle coatings, *Int. J. Heat Mass Transf.* 49 (5–6) (2006) 844–849.
- [19] C. Byon, S. Choi, S.J. Kim, Critical heat flux of bi-porous sintered copper coatings in FC-72, *Int. J. Heat Mass Transf.* 65 (2013) 655–661.
- [20] K.-H. Chu, R. Enright, E.N. Wang, Structured surfaces for enhanced pool boiling heat transfer, *Appl. Phys. Lett.* 100 (24) (2012) 241603.
- [21] I. Mudawar, A.H. Howard, C.O. Gersey, An analytical model for near-saturated pool boiling critical heat flux on vertical surfaces, *Int. J. Heat Mass Transf.* 40 (10) (1997) 2327–2339.
- [22] R. Patel, M. Meyer, J. Hartwig, I. Mudawar, Review of cryogenic pool boiling critical heat flux databases, assessment of models and correlations, and development of new universal correlations, *Int. J. Heat Mass Transf.* 190 (2022) 122579.
- [23] W. Bailey, E. Young, C. Beduz, Y. Yang, Pool boiling study on candidature of pentane, methanol and water for near room temperature cooling, in: Thermal and Thermomechanical Proceedings 10th Intersociety Conference on Phenomena in Electronics Systems, 2006, IEEE, 2006, pp. 599–603. ITHERM 2006.
- [24] C.-K. Guan, J.F. Klausner, R. Mei, A new mechanistic model for pool boiling CHF on horizontal surfaces, *Int. J. Heat Mass Transf.* 54 (17–18) (2011) 3960–3969.
- [25] N. Zuber, Hydrodynamic aspects of boiling heat transfer, United States Atomic Energy Commission, Technical Information Service, 1959.
- [26] J. Lienhard, V. Dhir, Hydrodynamic prediction of peak pool-boiling heat fluxes from finite bodies, (1973).
- [27] M.S. El-Genk, M. Pourghasemi, Subcooled boiling critical heat flux of HFE-7000 dielectric liquid on inclined rough Cu, *Int. J. Heat Mass Transf.* 175 (2021) 121354.
- [28] Z. Guo, M.S. El-Genk, An experimental study of saturated pool boiling from downward facing and inclined surfaces, *Int. J. Heat Mass Transf.* 35 (9) (1992) 2109–2117.
- [29] I. Vishnev, Effect of orienting the hot surface with respect to the gravitational field on the critical nucleate boiling of a liquid, *J. Eng. Phys.* 24 (1) (1973) 43–48.
- [30] M.S. El-Genk, H. Bostanci, Saturation boiling of HFE-7100 from a copper surface, simulating a microelectronic chip, *Int. J. Heat Mass Transf.* 46 (10) (2003) 1841–1854.
- [31] M. Arik, A. Bar-Cohen, Ebullient cooling of integrated circuits by Novec fluids, in: Proc. Pacific Rim Intersociety, Electronics Packaging Conference, 2001, pp. 18–23.
- [32] M.J. Brusstar, H. Merte Jr, Effects of buoyancy on the critical heat flux in forced convection, *J. Thermophys. Heat Transf.* 8 (2) (1994) 322–328.
- [33] M.J. Brusstar, H. Merte Jr, Effects of heater surface orientation on the critical heat flux—II. A model for pool and forced convection subcooled boiling, *Int. J. Heat Mass Transf.* 40 (17) (1997) 4021–4030.
- [34] V. Dhir, J. Linehard, Extended hydrodynamic theory of the peak and minimum pool boiling heat fluxes, (1973).
- [35] J. Li, Y. Huang, Y. Qiu, S. Wang, Q. Yang, K. Wang, Y. Zhu, Prediction of critical heat flux using different methods: a review from empirical correlations to the cutting-edge machine learning, *Int. Commun. Heat Mass Transf.* 160 (2025) 108362.
- [36] Z. Wang, X. Sun, M. Baghoolizadeh, S. Esmaeili, T. Alkhalifah, R. Marzouki, Review of pool boiling improvement with additives and nanofluids utilizing artificial intelligence, *Int. Commun. Heat Mass Transf.* 162 (2025) 108659.
- [37] F. Ahmad, M. Meyer, J. Hartwig, I. Mudawar, Development of new correlations for critical heat flux point wall temperature and transition boiling under steady-state saturated pool boiling of cryogenics, *Int. J. Heat Mass Transf.* 239 (2025) 126592.
- [38] F. Ahmad, D. Foster, S. Kim, M. Meyer, J. Hartwig, I. Mudawar, Method for generating a complete saturated pool boiling curve for cryogenic fluids under terrestrial gravity, *Int. J. Heat Mass Transf.* 243 (2025) 126876.
- [39] U. Sajjad, I. Hussain, C.-C. Wang, A high-fidelity approach to correlate the nucleate pool boiling data of roughened surfaces, *Int. J. Multiph. Flow* 142 (2021) 103719.
- [40] S. Mehdi, M. Borumand, G. Hwang, Accurate and robust predictions of pool boiling heat transfer with micro-structured surfaces using probabilistic machine learning models, *Int. J. Heat. Mass Transf.* 226 (2024) 125487.
- [41] U. Sajjad, S. Mehdi, I. Hussain, T.-u. Rehman, M. Sultan, M.M. Rashidi, W.-M. Yan, Physics driven interpretable deep learning-based insights into boiling crisis of smooth and roughened surfaces, *Alex. Eng. J.* 116 (2025) 112–128.
- [42] H. Noh, S. Lee, S.-M. Kim, I. Mudawar, Utilization of XGBoost algorithm to predict dryout incipience quality for saturated flow boiling in mini/micro-channels, *Int. J. Heat Mass Transf.* 231 (2024) 125827.
- [43] I. Mudawar, S.J. Darges, V. Devahdhanush, M.M. Hasan, H.K. Nahra, R. Balasubramaniam, J.R. Mackey, Pressure drop characteristics and prediction techniques (models/correlations and artificial neural networks) for microgravity flow boiling onboard the International Space Station, *Int. J. Heat Mass Transf.* 240 (2025) 126593.
- [44] U. Sajjad, W.-M. Yan, I. Hussain, S. Mehdi, M. Sultan, H.M. Ali, Z. Said, C.-C. Wang, Physics and correlations informed deep learning to foresee various regimes of the pool boiling curve, *Eng. Appl. Artif. Intell.* 136 (2024) 108867.
- [45] U. Sajjad, Y.-H. Chu, H. Yaqoob, A. Sengupta, H.M. Ali, K. Hamid, W.-M. Yan, Physics-based parameters selection and machine learning driven prediction of pool boiling bubble departure diameter, *Energy Convers. Manag.* X 25 (2025) 100795.
- [46] J. Kim, J. Yeom, D. Kong, H. Lee, S.-m. Kim, Analysis of the enhancing mechanism in pool boiling heat transfer through wetting speed for rough aluminum surfaces with FC-72, *Int. J. Heat Mass Transf.* 150 (2020) 119325.
- [47] M. Al Masri, S. Cioulachtjian, C. Veillas, I. Verrier, Y. Jourlin, J. Ibrahim, M. Martin, C. Pupier, F. Lefevre, Nucleate boiling on ultra-smooth surfaces: explosive incipience and homogeneous density of nucleation sites, *Exp. Therm. Fluid Sci.* 88 (2017) 24–36.
- [48] N. Tran, U. Sajjad, R. Lin, C.-C. Wang, Effects of surface inclination and type of surface roughness on the nucleate boiling heat transfer performance of HFE-7200 dielectric fluid, *Int. J. Heat Mass Transf.* 147 (2020) 119015.
- [49] J.P. McHale, S.V. Garimella, Bubble nucleation characteristics in pool boiling of a wetting liquid on smooth and rough surfaces, *Int. J. Multiph. Flow* 36 (4) (2010) 249–260.
- [50] J. Kim, S. Jun, J. Lee, J. Godinez, S.M. You, Effect of surface roughness on pool boiling heat transfer of water on a superhydrophilic aluminum surface, *J. Heat Transf.* 139 (10) (2017).
- [51] C. Bombardieri, C. Manfretti, Influence of wall material on nucleate pool boiling of liquid nitrogen, *Int. J. Heat Mass Transf.* 94 (2016) 1–8.
- [52] B. Bon, C.-K. Guan, J.F. Klausner, Heterogeneous nucleation on ultra smooth surfaces, *Exp. Therm. Fluid Sci.* 35 (5) (2011) 746–752.
- [53] P.F. Alvarino, M.L.S. Simón, M. dos Santos Guzella, J.M.A. Paz, J.M.S. Jabardo, L. C. Gómez, Experimental investigation of the CHF of HFE-7100 under pool boiling conditions on differently roughened surfaces, *Int. J. Heat Mass Transf.* 139 (2019) 269–279.
- [54] S. Ahmad, T. Karayiannis, D. Kenning, A. Luke, Compound effect of EHD and surface roughness in pool boiling and CHF with R-123, *Appl. Therm. Eng.* 31 (11–12) (2011) 1994–2003.
- [55] S. Fan, L. Jiao, K. Wang, F. Duan, Pool boiling heat transfer of saturated water on rough surfaces with the effect of roughening techniques, *Int. J. Heat Mass Transf.* 159 (2020) 120054.
- [56] M.R.M. Arenales, S. Kumar, L.-S. Kuo, P.-H. Chen, Surface roughness variation effects on copper tubes in pool boiling of water, *Int. J. Heat Mass Transf.* 151 (2020) 119399.
- [57] A. Suszko, M.S. El-Genk, Saturation boiling of PF-5060 on rough Cu surfaces: bubbles transient growth, departure diameter and detachment frequency, *Int. J. Heat Mass Transf.* 91 (2015) 363–373.
- [58] S.J. Thiagarajan, R. Yang, C. King, S. Narumanchi, Bubble dynamics and nucleate pool boiling heat transfer on microporous copper surfaces, *Int. J. Heat Mass Transf.* 89 (2015) 1297–1315.
- [59] J. Kim, S. Jun, R. Lakshminarain, S.M. You, Effect of surface roughness on pool boiling heat transfer at a heated surface having moderate wettability, *Int. J. Heat Mass Transf.* 101 (2016) 992–1002.
- [60] M.S. El-Genk, M. Pourghasemi, Experimental investigation of saturation boiling of HFE-7000 dielectric liquid on rough copper surfaces, *Therm. Sci. Eng. Prog.* 15 (2020) 100428.
- [61] J.P. McHale, S.V. Garimella, Nucleate boiling from smooth and rough surfaces—Part 1: fabrication and characterization of an optically transparent heater–sensor substrate with controlled surface roughness, *Exp. Therm. Fluid Sci.* 44 (2013) 456–467.
- [62] J.P. McHale, S.V. Garimella, Nucleate boiling from smooth and rough surfaces—Part 2: analysis of surface roughness effects on nucleate boiling, *Exp. Therm. Fluid Sci.* 44 (2013) 439–455.
- [63] C.M. Kruse, T. Anderson, C. Wilson, C. Zuhlke, D. Alexander, G. Gogos, S. Ndao, Enhanced pool-boiling heat transfer and critical heat flux on femtosecond laser processed stainless steel surfaces, *Int. J. Heat Mass Transf.* 82 (2015) 109–116.
- [64] G. Ribatski, J.M.S. Jabardo, Experimental study of nucleate boiling of halocarbon refrigerants on cylindrical surfaces, *Int. J. Heat Mass Transf.* 46 (23) (2003) 4439–4451.
- [65] R. Benjamin, A. Balakrishnan, Nucleation site density in pool boiling of saturated pure liquids: effect of surface microroughness and surface and liquid physical properties, *Exp. Therm. Fluid Sci.* 15 (1) (1997) 32–42.
- [66] B.D. Bock, M. Bucci, C.N. Markides, J.R. Thome, J.P. Meyer, Pool boiling of refrigerants over nanostructured and roughened tubes, *Int. J. Heat Mass Transf.* 162 (2020) 120387.
- [67] B.D. Bock, J.P. Meyer, J.R. Thome, Falling film boiling and pool boiling on plain circular tubes: influence of surface roughness, surface material and saturation temperature on heat transfer and dryout, *Exp. Therm. Fluid Sci.* 109 (2019) 109870.
- [68] A. Bode, Single bubble dynamics and transient pressure during subcooled nucleate pool boiling, *Int. J. Heat Mass Transf.* 51 (1–2) (2008) 353–360.
- [69] E.I. Eid, R.A. Khalaf-Allah, M. Tolan, Enhancement of pool boiling characteristics by an addition of nano aluminum oxide to R-141b over a rough horizontal steel circular heater, *Int. J. Refrig.* 98 (2019) 311–322.
- [70] K. Ferjančić, I. Golobić, Surface effects on pool boiling CHF, experimental thermal and fluid science, 25(7) (2002) 565–571.
- [71] P. Goel, A.K. Nayak, P.P. Kulkarni, J.B. Joshi, Experimental study on bubble departure characteristics in subcooled nucleate pool boiling, *Int. J. Multiph. Flow* 89 (2017) 163–176.

- [72] J.M.S. Jabardo, G. Ribatski, E. Stelute, Roughness and surface material effects on nucleate boiling heat transfer from cylindrical surfaces to refrigerants R-134a and R-123, *Exp. Therm. Fluid Sci.* 33 (4) (2009) 579–590.
- [73] M.-G. Kang, Effect of surface roughness on pool boiling heat transfer, *Int. J. Heat Mass Transf.* 43 (22) (2000) 4073–4085.
- [74] J.S. Kim, A. Girard, S. Jun, J. Lee, S.M. You, Effect of surface roughness on pool boiling heat transfer of water on hydrophobic surfaces, *Int. J. Heat Mass Transf.* 118 (2018) 802–811.
- [75] S.H. Kim, G.C. Lee, J.Y. Kang, H.S. Park, M.H. Kim, A study of nucleate bubble growth on microstructured surface through high speed and infrared visualization, *Int. J. Multiph. Flow* 95 (2017) 12–21.
- [76] I.S. Kiyomura, T.S. Mogaji, L.L. Manetti, E.M. Cardoso, A predictive model for confined and unconfined nucleate boiling heat transfer coefficient, *Appl. Therm. Eng.* 127 (2017) 1274–1284.
- [77] A. Luke, Pool boiling heat transfer from horizontal tubes with different surface roughness, *Int. J. Refrig.* 20 (8) (1997) 561–574.
- [78] A. Luke, Active and potential bubble nucleation sites on different structured heated surfaces, *Chem. Eng. Res. Des.* 82 (4) (2004) 462–470.
- [79] M. Mpholo, T. Mathaba, H.H. Bau, A 2D analysis of surface roughness for prediction of boiling incipience, *Int. J. Heat Mass Transf.* 53 (7–8) (2010) 1313–1318.
- [80] J.L. Parker, M.S. El-Genk, Enhanced saturation and subcooled boiling of FC-72 dielectric liquid, *Int. J. Heat Mass Transf.* 48 (18) (2005) 3736–3752.
- [81] I. Pioro, W. Rohsenow, S. Doerffer, Nucleate pool-boiling heat transfer. I: review of parametric effects of boiling surface, *Int. J. Heat Mass Transf.* 47 (23) (2004) 5033–5044.
- [82] Y. Qi, J.F. Klausner, R. Mei, Role of surface structure in heterogeneous nucleation, *Int. J. Heat Mass Transf.* 47 (14–16) (2004) 3097–3107.
- [83] D. Sarker, R. Franz, W. Ding, U. Hampel, Single bubble dynamics during subcooled nucleate boiling on a vertical heater surface: an experimental analysis of the effects of surface characteristics, *Int. J. Heat Mass Transf.* 109 (2017) 907–921.
- [84] A. Suszko, M.S. El-Genk, Dielectric liquids natural convection on small rough Cu surfaces at different orientations, *Int. J. Heat Mass Transf.* 81 (2015) 289–296.
- [85] A. Swain, S. Swain, S.P. Kar, R. Sarangi, P. Chandrasekhar, Effect of surface characterization on the application of pool boiling heat transfer augmentation, *Mater. Today: Proc.* 38 (2021) 469–473.
- [86] A. Walunj, A. Sathyabhama, Transient CHF enhancement in high pressure pool boiling on rough surface, *Chem. Eng. Process.-Process Intensif.* 127 (2018) 145–158.
- [87] J.-Y. Zhang, L.-W. Fan, J.-Q. Li, Z.-T. Yu, Surface-roughness-boosted critical heat flux enhancement during quenching boiling on wicking surfaces, *Int. J. Heat Mass Transf.* 162 (2020) 120364.